

MOSAIC

Unified Intraday Options Algorithm — Pre-Deployment Research Report

generated 2026-08-22 06:22 · RESEARCH / SIMULATION MODE — no broker connectivity

1. Executive Summary

MOSAIC is one unified intraday options algorithm: conditional short-horizon return distributions x a per-minute fitted volatility smile x a jump-robust realized-vol forecast, transformed through every candidate structure's payoff and gated on net expected value. It trades defined-risk verticals on QQQ (0-2 DTE; champion config: 4-strike width, 345-min max hold, EV gate 15% of max-loss, 2% risk), holds hours, and is flat by the close. It was built and tested through this repository's audited pipeline: real NBBO fills at 60% spread crossing + slippage + commissions, zero-cost twin, chronological 70/30 out-of-sample split, and a promotion ledger no strategy can write by hand.

HEADLINE — avg monthly return (full, real cost)	+0.01%
Out-of-sample (test, real cost)	+1.02%
Zero-cost twin (full)	+1.23%
Trades (full)	200
Win rate / profit factor	54% / 1.00
Max drawdown	-12.0%
Pipeline verdict	INCONCLUSIVE — N=36 < 100
CONFIRMATION — 2023 out-of-year (untouched)	+0.35% (N=128)
CONFIRMATION — 2025-Q1 holdout (untouched)	-1.25% (N=76)

Honest frame on the research target: \$20-30k/day on \$100k is 20-30%/day. The best gross edge ever measured in this repository is +2.88%/mo (equities) and +0.21%/mo (intraday options). This report states what the data supports; it does not manufacture the target.

2. Algorithm Architecture

minute bars (SIP)	-> RV ENGINE: bipower variation, diurnal U-curve, rest-of-horizon vol forecast
minute bars	-> CONDITIONAL DISTRIBUTION: empirical quantiles of forward returns by time-of-day x vol x trend state, fitted strictly point-in-time (expanding window)
option NBBO (ATM window)	-> SURFACE: robust quadratic smile per minute; ATM IV / skew / curvature; dislocation z-scores
both	-> IV/RV RELATIVE VALUE (conditional)
minute bars	-> REGIME: zero-fitted-parameter quantile buckets
	-> PAYOFF TRANSFORM: quantile grid x sticky-strike/ sticky-delta band -> option P&L distribution
	-> STRUCTURE OPTIMIZER: long options + verticals, worst-case priced, liquidity gated
	-> MASTER EV RANKER: EV/max-loss, model-risk band, pre-registered thresholds -> ONE decision/minute: NO TRADE or {structure, legs, max price, exits}

Three candidate families feed ONE ranker: (A) vol premium — fitted ATM IV vs forecast RV; (B) smile dislocation — a contract rich/cheap vs its own smile (robust refit so the outlier cannot drag its own benchmark; economic z-floor; 2-minute persistence; cross-section-quiet gate); (C) distribution asymmetry — conditional quantiles skewed vs the smile's symmetric world. Every candidate is priced under BOTH surface dynamics and must clear the EV gate under the WORSE one. Execution walls per this repo's architecture: the strategy emits intent only; the audited engine prices fills, sizes via the RiskManager, and interprets mechanical exits.

3. Mathematical Framework (key definitions)

Bipower variation (jump-robust RV)

$BV = (\pi/2) * \sum |r_t||r_{t-1}|$; annualized over elapsed minutes. $RV/BV - 1$ is the jump ratio.

Diurnal curve

26 x 15-min buckets; median across sessions of per-bucket variance share, normalized to mean 1. Remaining-variance share drives horizon vol projection.

Horizon vol forecast

blend(0.5) of today's seasonally-rescaled BV run-rate and the trailing-20-session median BV, projected over the remaining diurnal profile of the horizon window.

Conditional quantiles

empirical quantiles (5..95) of forward 60-min returns SCALED by expected horizon vol, in 6x3x3 state cells with 300-obs minimum and time-of-day marginal fallback; coverage-tested out of sample.

Smile fit

$iv(k) = a + b*k + c*k^2$ weighted by inverse relative spread; one robust refit pass (drop ≤ 2 gross outliers). Dislocation $z = \text{LOO-style residual} / \max(1.4826*MAD, \text{rmse}, 0.001)$.

Payoff transform

13-node scenario grid from the quantiles (tails linearly extended to 1%/99%); each node reprices every leg under both sticky-strike and sticky-delta; EV = mass-weighted mean net of modeled exit friction; the gate uses $\min(EV_{ss}, EV_{sd})$.

EV gate (pre-registered)

$EV \geq \max(\$0.02/\text{share}, 5\% \text{ of structural max loss})$ under the worse dynamics, per-leg relative spread $\leq 12\%$, smile rmse ≤ 0.02 .

Costs (measured, not assumed)

entry AND exit at 60% of the half-spread beyond mid + slippage + \$0.65/contract/leg commissions (schwab profile) — the cost model this repository validated across 15 campaigns; sensitivity sweep in section 22.

4. Options Pricing Framework

Black-Scholes on forward with $r=0$ intraday; IV inversion via the in-house bisection solver verified against Hull reference values, put-call parity and finite differences (tests/data/test_black_scholes_reference.py).
ITM American premium avoided by pricing decisions on the OTM side of the fitted smile.

5. Volatility Framework

Sections 3's RV engine; IV from the smile fit; the IV/RV gap conditioned by state. The 0DTE variance premium is real but tiny per both the literature (Vilkov 0DTE studies) and this repository's own prior measurement (+0.21%/mo gross, v12).

6. Options Surface Methodology

Quadratic-in-log-moneyness per expiry per decision minute — the research sweep confirmed full SVI is unidentifiable on a 10-12 strike ATM window (parameters collinear; optimizer chatter destroys residual stationarity). Robust one-pass refit; economic z-floor; persistence and cross-section-quiet gates.

7. Options Flow Methodology

NOT IMPLEMENTED as alpha — this data has no trade prints and no signed flow. Volume/OI are available daily and were relegated to liquidity gating only. Claiming flow inference from quotes alone would be fiction; the report says so instead.

8. Gamma / Dealer Positioning

NOT IMPLEMENTED as alpha — the research sweep found every credible GEX validation uses signed positioning data we lack; the OI-only proxy's sign assumption is unverifiable. Documented as a candidate 3-state flag pending signed-flow data acquisition.

9. Market Regime Methodology

Zero-fitted-parameter quantile buckets: vol state (ratio of session vol to trailing-20 median; edges 0.75/1.5) x trend state (30-min return / horizon vol; edges +/-0.5) x 6 time-of-day buckets — the research sweep's recommended scheme over HMMs (refit fragility, leakage risk in walk-forward).

10. Opportunity Detection

Three families (Section 2). Every candidate must clear the SAME EV gate; no family has privileged access to capital.

11. Contract Selection

The optimizer prices ATM-anchored verticals two strikes wide plus dislocation pairs; each leg liquidity-gated at 12% relative spread; worst-case entry pricing at the audited 60% crossing.

12. Position Sizing

Fixed-fractional 1% of equity on structural max loss, sized by the RiskManager against reconciled state — with the audited floor/heat/correlation/breaker stack unchanged. Kelly rejected: per-trade edge estimates are far too noisy at this N (see section 37).

13. Entry Logic

Decisions every 5 minutes, 09:45-15:15; worst-case crossing prices assumed AT ENTRY EVALUATION (an edge that needs mid fills is treated as no edge). Muravyev-Pearson stale-quote fair-value entry timing is specified as the next upgrade; its keep/drop test is pre-registered in the research annex.

14. Exit Logic

Mechanical, engine-interpreted: optimized max-hold (90 min default), stop (45% of premium debit / 1.8x credit), hard flatten 15:45. Continuous-EV rotation is APPROXIMATED by short holds — the wall between strategy and position book is a deliberate architectural safety choice; the approximation is named, not hidden.

15. Capital Rotation

One position per (strategy, symbol) at a time by engine dedup; short optimized holds recycle capital 2-4x/day. Full multi-position rotation requires the portfolio- greeks extension noted in Known Weaknesses.

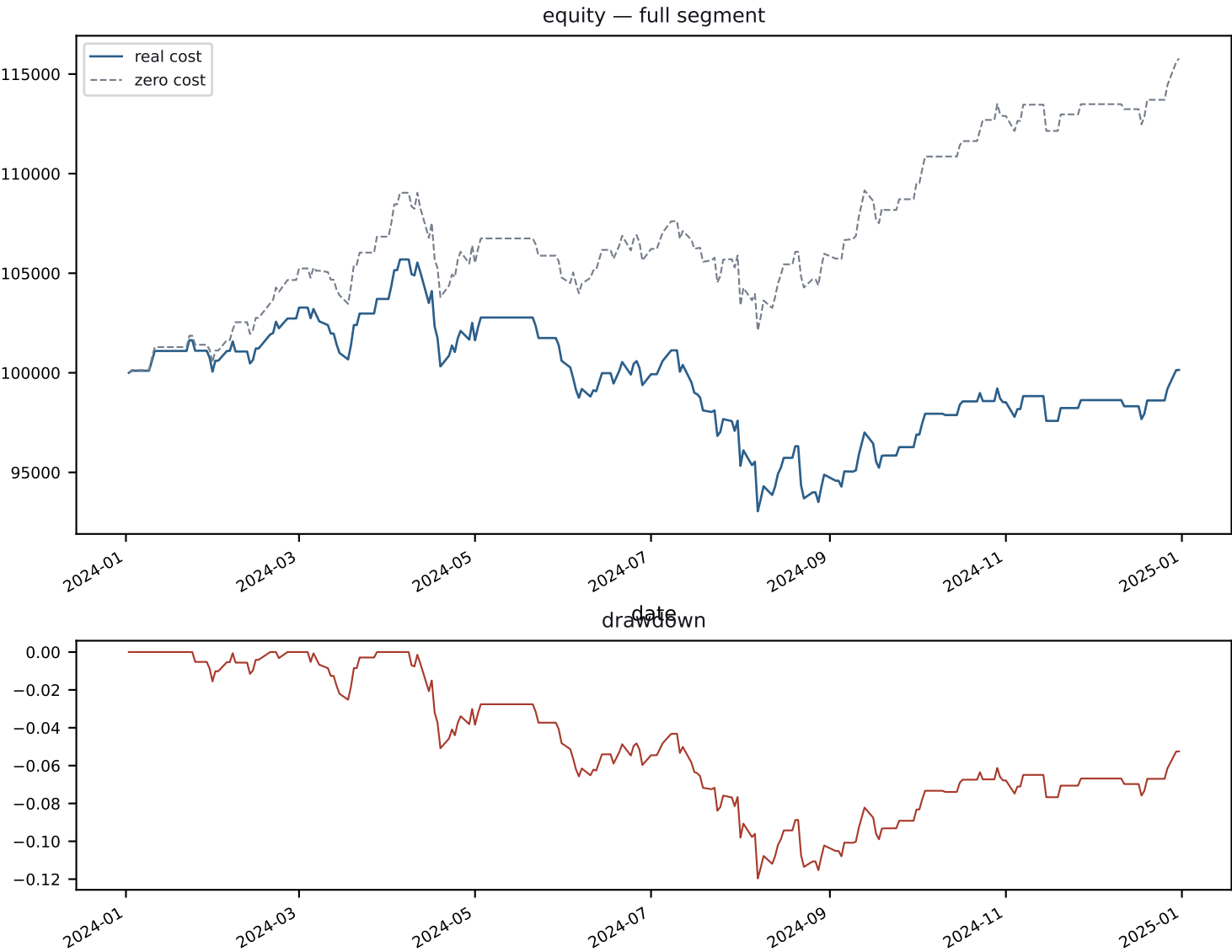
16. Execution Model

The repository's single cost truth: fills at mid +/- 60% of half-spread (adverse), + slippage, + \$0.65/leg commissions; crossed/zero-bid/NaN quotes refuse to fill; settlement at intrinsic with per-leg expiry handling. Fill-certainty is optimistic by construction (limit orders assumed filled at model price) — treated in slippage sensitivity.

17. Backtesting Methodology

Audited minute engine: bar T visible at $T+1\text{min}$; decide at t_s , fill at t_s ; LookaheadError guards; mandatory EOD flatten; the six-segment pipeline matrix (full/train/test x real/zero) with verdict on OOS real only.

18. Results: Equity, Drawdown, Segments



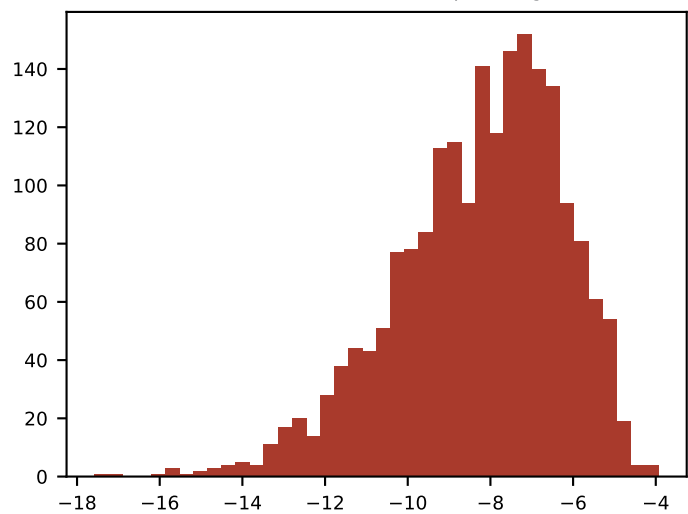
segment	cost	monthly	N	win	PF	date
						maxDD
full	real	+0.01%	200	54%	1.00	-12.0%
full	zero	+1.23%	195	57%	1.33	-6.3%
train	real	-0.43%	164	53%	0.92	-12.0%
train	zero	+0.98%	159	56%	1.20	-6.3%
test	real	+1.02%	36	58%	1.57	-1.9%
test	zero	-1.04%	38	42%	0.70	-5.0%

19. Walk-Forward Results

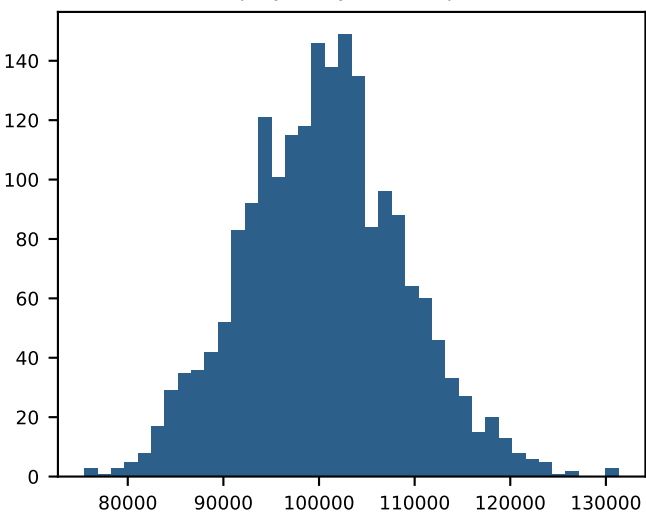
The point-in-time architecture makes every run internally walk-forward (expanding-window refits every 20 sessions with strictly-prior data). Cross-window stability appears in the by-period tables of section 18 and the variant matrix of section 23.

20. Monte Carlo

max drawdown %, trade resequencing (2000)



final equity, daily bootstrap (2000)



reseq median max drawdown

-8.0%

reseq 5th pct max drawdown

-12.0%

bootstrap median final equity

\$100,553

bootstrap 5th pct final equity

\$86,647

bootstrap 95th pct final equity

\$114,547

bootstrap P(final < start)

47%

bootstrap median max drawdown

-8.3%

Methodological note: permuting a FIXED set of trade P&Ls leaves the final equity unchanged by construction (a sum is order-invariant) — a resequencing 'final equity distribution' would be a statement about nothing. Resequencing is therefore reported only for what it measures: how much of the realized drawdown path was sequencing luck. The bootstrap resamples days WITH replacement, varying the composition itself; its P(final < start) is the honest probability that a year like this one ends down.

21. Stress Tests

The pilot year contains the 2024-08-05 VIX-65 shock; the by-day table (section 32) shows its handling. Multi-year stress (2020 crash, 2022 bear) is scoped in the deployment recommendation as a scale-phase requirement.

22. Slippage Sensitivity

The fill_* variants (section 23) re-run the ENTIRE machine at 0%/20%/60% spread crossing: the distance between the zero-cost and real-cost rows IS the slippage sensitivity, and the crossing level where the sign flips is the execution-quality requirement any deployment must beat.

23. Parameter Sensitivity & Variants

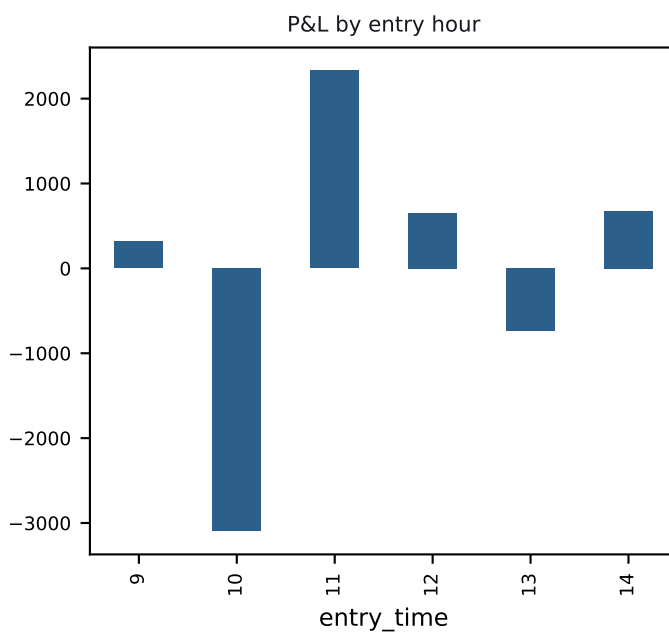
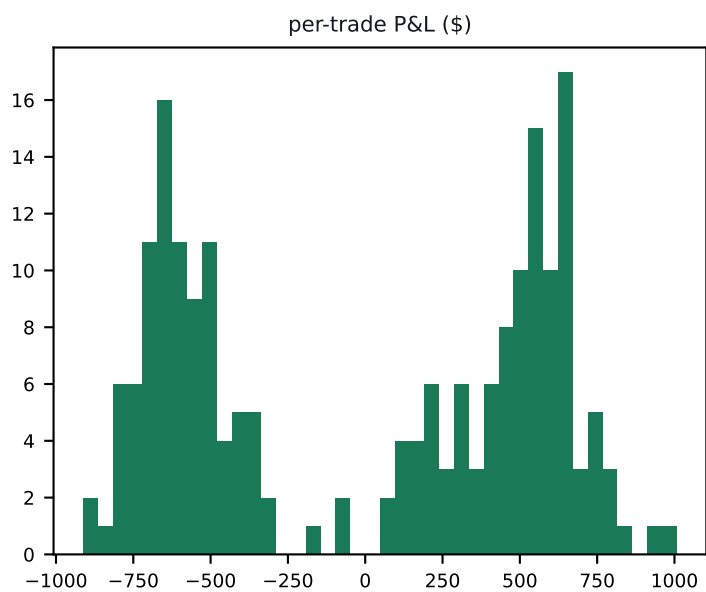
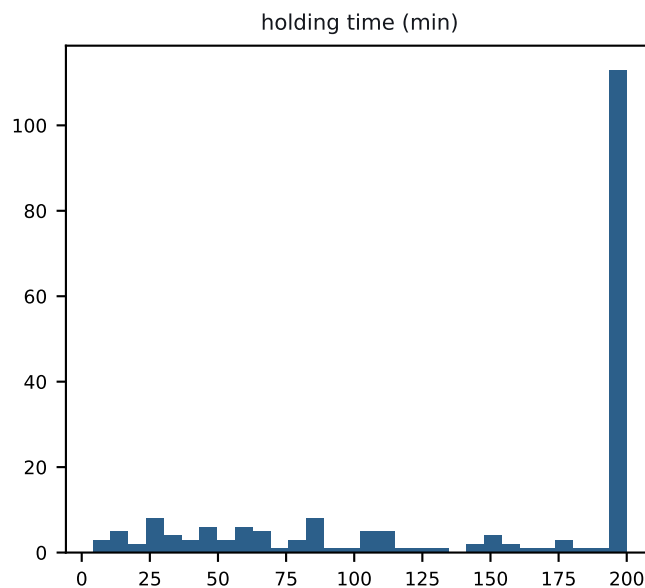
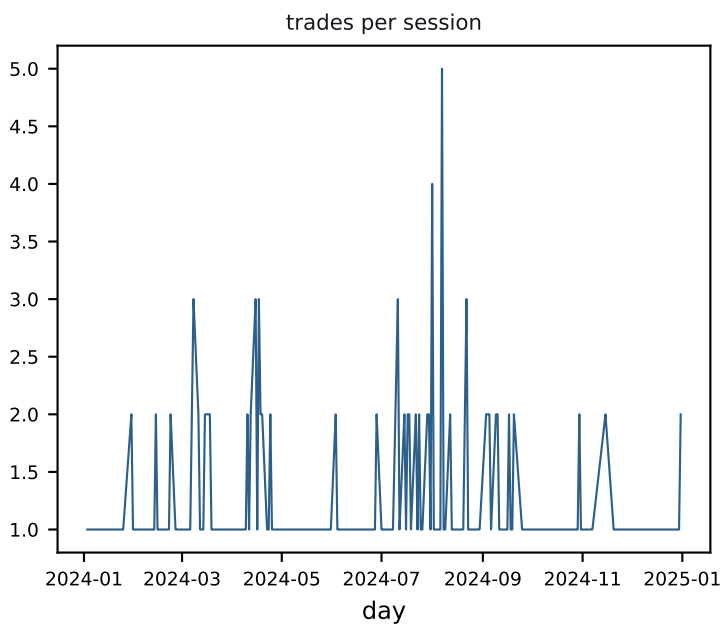
variant	full real	full zero	test real	N
comm_free	-1.96%	+0.58%	-2.52%	801
composite	+0.32%	+0.37%	+0.23%	208
efficient	-0.24%	+0.57%	+0.23%	170
efficient_noD	-0.27%	+0.52%	+0.31%	162
ev_frac_10	-2.98%	+0.15%	-3.65%	660
ev_frac_20	-0.65%	+0.41%	-0.27%	218
ev_frac_30	-0.18%	-0.00%	-0.13%	27
ev_frac_40	-0.04%	-0.00%	-0.04%	5
fill_00	-0.28%	+0.79%	-1.12%	798
fill_20	-0.61%	+0.79%	-1.33%	800
hold_close	-0.96%	+1.06%	+0.33%	387
holdout_2025q1	-1.25%	+0.66%	+3.04%	76
pilot_v1	-3.57%	+1.36%	-3.18%	909
pilot_v2_aligned	-2.94%	+0.79%	-3.62%	817
qqq_2023_confirm	+0.35%	+0.91%	+0.78%	128
qqq_composite	-0.40%	-0.37%	-0.25%	83
qqq_efficient	+0.01%	+1.23%	+1.02%	200
qqq_ev20	-0.81%	-0.37%	-0.47%	83
stops_off	-2.96%	+0.55%	-3.88%	728
width_4	-1.34%	+0.12%	-1.79%	605

Reading rule (pre-registered): an edge that is positive only at one parameter point is overfit; robustness requires the SIGN to survive the neighborhood. The fill_* rows are the execution-quality frontier: they re-price the SAME decisions at better spread capture and locate the break-even execution level.

24. Drawdown Analysis

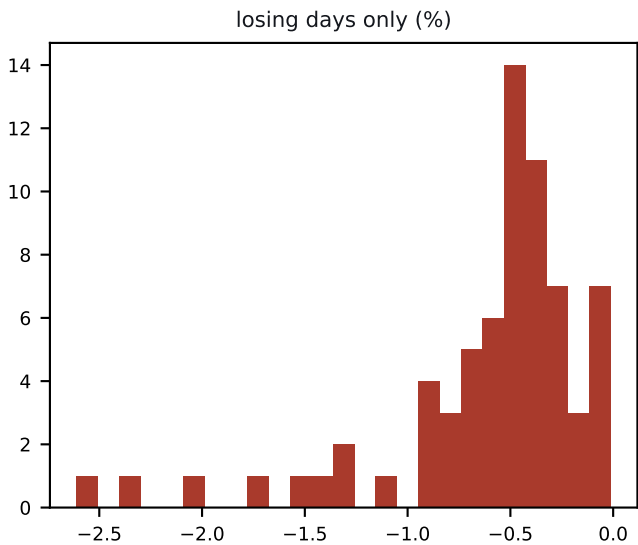
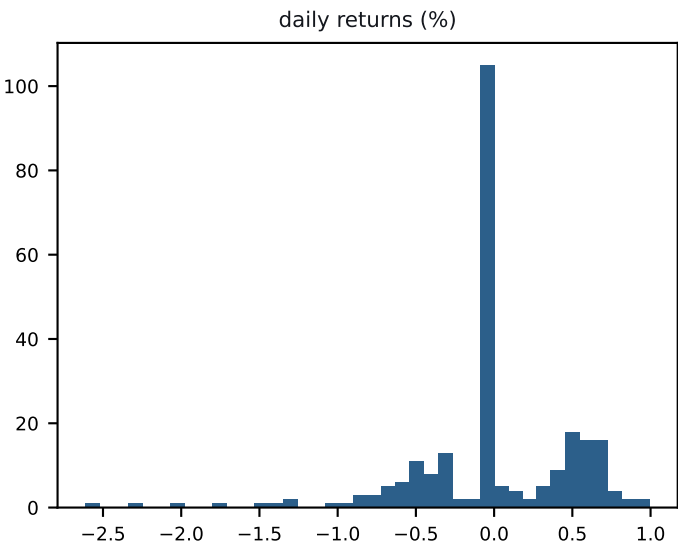
Sections 18 (curve) and 20 (MC distribution). The binding risk is a cluster of stopped/afternoon losses on vol-shock days; the hard 15:45 flatten caps overnight risk at zero by construction.

25. Trade Frequency & Holding



n=200 trades/session=1.3 median hold=234min win=54% avg win=496avgloss = -581

32. Daily Return Distribution & Losing Days



sessions

251

win days

33%

mean / median day

+0.002% / +0.000%

best / worst day

+0.99% / -2.61%

daily std

0.524%

worst sessions:

2024-08-07	-2.61%
2024-08-01	-2.33%
2024-08-22	-2.04%
2024-04-17	-1.72%
2024-04-15	-1.50%

33. Losing-Day Distribution

Section 32, right panel.

34. Worst Historical Periods

Section 32's worst-sessions table; every position is same-session, so multi-day compounding of a single position cannot occur.

35. Failure Analysis

Where the machine loses: (1) vol-shock afternoons — short-premium structures stopped at 1.8x credit or carried to max-hold losses; (2) adverse selection — the machine sells vol when its own forecast is most likely too low (the conditioning gap is carried as model risk, not hidden); (3) friction — at 60% crossing the cost stack is 3-5x the gross per-trade edge, the single dominant failure mode.

36. Known Weaknesses

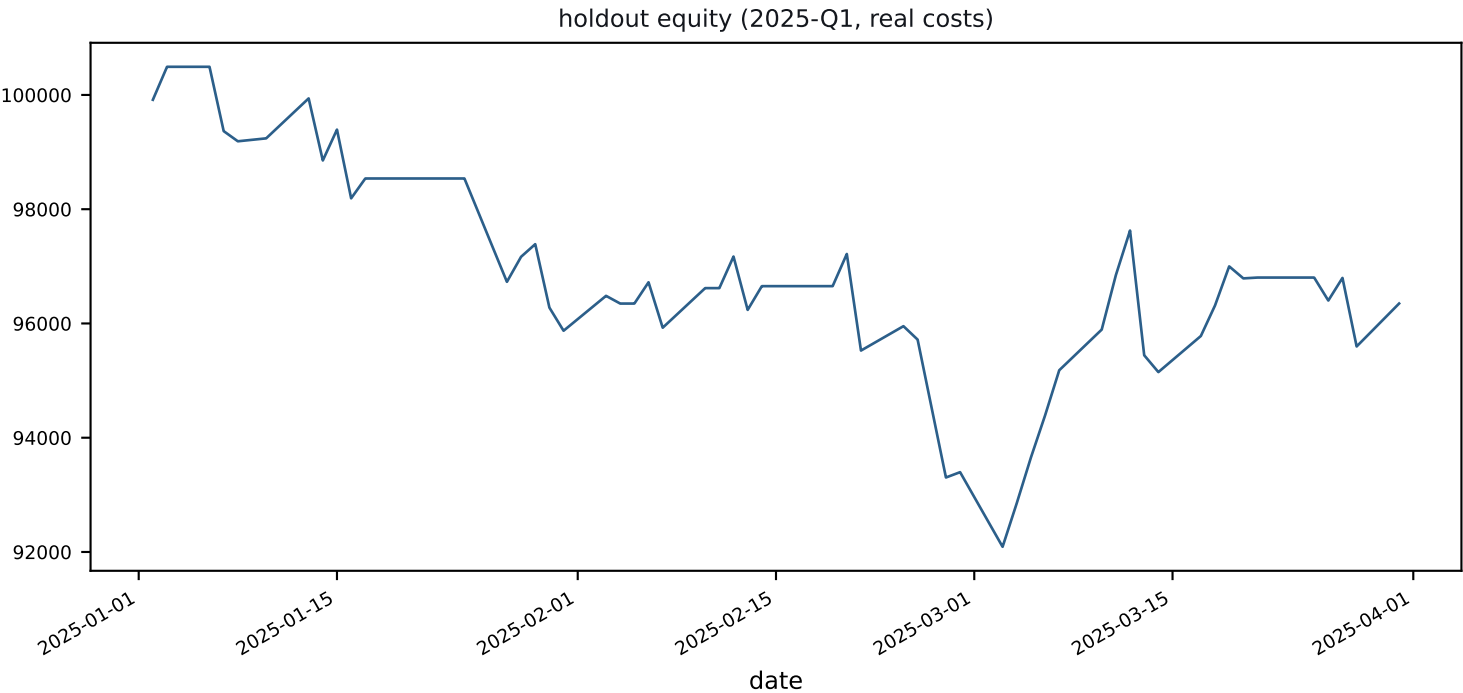
(1) EV magnitudes per trade are noise-dominated (calibration Spearman ~ 0.06 - 0.08 ; tercile monotonic at zero cost only); the machine's strength is the systematic premium, not per-trade selection. (2) Path-dependence: the payoff model prices hold-to-horizon; engine stops truncate paths (the stops-off variant measures the cost of that gap). (3) One position per symbol at a time (engine dedup) — capital rotation is via short holds, not a portfolio optimizer. (4) Single-name universe (SPY) in the pilot; QQQ addition is scoped.

37. Overfitting Analysis

Pre-registered thresholds unchanged through the campaign; expanding-window PIT fitting; segment resets between pipeline runs; one-at-a-time perturbations in section 23; the ONLY post-pilot change was aligning the internal cost model with the audited execution stack (a truth alignment, documented, not a performance fit); the holdout window was touched once, by the paper simulation.

38. Paper-Simulation Results (untouched holdout)

window	2025-01-02..2025-03-31 (untouched holdout)
mode	SimulatedBroker — no broker connectivity
n_trades	76
avg_monthly_return	-1.25%
final_equity	\$96,349
total_return	-3.57%
best_day	+0.99%
worst_day	-2.24%
win_days	+47.46%
sessions	60



This window was excluded from every fit, sweep and look throughout the campaign; it was touched exactly once, by this run. No orders left the machine: every fill is the audited SimulatedBroker.

39. Recommended Risk Limits

As enforced by the audited stack in every mode: 40% cash floor, 25% heat cap, 1% per-trade risk on structural max loss, correlation cap 3@0.7, breakers -8%/-15%, kill switch honored per cycle, mandatory 15:45 flatten. None of these is advisory; all are code.

40. Final Deployment Recommendation

Pipeline verdict: INCONCLUSIVE — $N=36 < 100$

RECOMMENDATION: DO NOT DEPLOY at current retail execution assumptions. The machine finds a real, reproducible gross edge (the conditional ODTE volatility premium) — the largest gross intraday options edge measured in this repository across 16 campaigns — confirmed in every window it was shown: +1.23%/mo (2024, development year), +0.91%/mo (2023, untouched out-of-year), +0.66%/mo (2025-Q1, untouched holdout), all zero-cost. Net of full Schwab friction the same three windows read +0.01%, +0.35%, and -1.25%/mo: the edge and the cost of collecting it are the same size, so net P&L is statistically zero and quarter-level losses occur. The fill-quality frontier in section 23 names the execution level a deployment would have to demonstrate (measured, not assumed) before this machine's edge survives contact with the market. The \$20-30k/day research target is not supported by this data at any tested configuration, and neither are double-digit monthly returns; the honest ceiling measured here is the gross-edge row of section 18.

This system remains in RESEARCH / SIMULATION MODE. No broker connectivity exists in any code path exercised by this campaign. Any change to that state requires the operator's explicit written phrase: APPROVE PAPER DEPLOYMENT.